INCH-POUND

MIL-DTL-46085C
27 May 2025
SUPERSEDING
MIL-R-46085B
24 April 1984

DETAIL SPECIFICATION

RHODIUM PLATING, ELECTRODEPOSITED

REACTIVATED AFTER 27 MAY 2025 AND MAY BE USED FOR NEW AND EXISTING DESIGN AND ACQUISITIONS.

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers the requirements for the electrodeposition of rhodium over metallic surfaces. (see 6.1).

1.2 Classification. Rhodium plating should be applied under the following classes, as specified (see 6.2 and 6.6):

Class 1 - 0.000002 inch (0.0508 micrometers) thick, minimum rhodium coating
Class 2 - 0.00001 inch (0.2540 micrometers) thick, minimum rhodium coating
Class 3 - 0.00002 inch (0.5080 micrometers) thick, minimum rhodium coating
Class 4 - 0.00010 inch (2.5400 micrometers) thick, minimum rhodium coating
Class 5 - 0.00025 inch (6.3500 micrometers) thick, minimum rhodium coating

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory (ARL), Army Research Directorate, Specifications & Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product Attributes

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes
ISO 9000	-	Quality Management Systems - Fundamentals and Vocabulary

(Copies of this document are available online at https://ansi.org/).

ASTM INTERNATIONAL

ASTM B244	-	Standard Test Method for Measurement of Thickness of Anodic Coatings on Aluminum and of Other Nonconductive Coatings on Nonmagnetic Basis Metals with Eddy-Current Instruments
ASTM B487	-	Standard Test Method for Measurement of Metal and Oxide Coating Thickness by Microscopical Examination of Cross Section
ASTM B499	-	Standard Test Method for Measurement of Coating Thicknesses by the Magnetic Method: Nonmagnetic Coatings on Magnetic Basis Metals

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ASTM B567	-	Standard Test Method for Measurement of Coating Thickness by the Beta Backscatter Method
ASTM B568	-	Standard Test Method for Measurement of Coating Thickness by X-Ray Spectrometry

(Copies of these documents are available online at https://www.astm.org).

SAE INTERNATIONAL STANDARDS / Aerospace Material Specifications

AMSQQN290	-	Nickel Plating (Electrodeposited)

(Copies of these documents are available from https://www.sae.org).

2.4 Order of precedence. Unless otherwise noted herein or within the contract or purchase order, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Preproduction sample. When specified in the invitation for bids, contract or order (see 6.2), samples of electrodeposited rhodium, representative of the class to be furnished in accordance with this specification, shall be subject to preproduction inspection and approval prior to submission of lots for acceptance inspection.

3.2 Materials. The material used shall produce deposits which meet the requirements of this specification.

3.3 General requirements.

3.3.1 Plating application. Unless otherwise specified in the contract or purchase order, the plating shall be applied after all machining, brazing, welding, forming, and perforating of the article have been completed.

3.3.2 Stress relief treatment. All ferrous parts having a hardness of Rockwell C33 and above which are machined, ground, cold formed or cold straightened shall be given a heat treatment at a minimum of 190 ± 14 °C (375 °F) for 3 hours or more prior to cleaning and plating for the relief of damaging residual tensile stresses.

3.3.3 Embrittlement relief. All ferrous parts having a hardness of Rockwell C40 and higher shall be baked at a minimum of 190 ± 14 °C (375 °F) for 3 hours or more, within 4 hours after plating to provide hydrogen embrittlement relief. The baked parts. when tested as specified in 4.5.4, shall not crack or fail by fracture. Plated springs and other parts subject to flexure shall not be flexed prior to hydrogen embrittlement relief treatment.

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3.4 Preparation of surfaces for plating. Surfaces shall be electrocleaned or otherwise subjected to any cleaning process necessary to ensure rhodium plating meeting the requirements specified herein. Unless specified otherwise, ferrous metals having a hardness of Rockwell C40 or higher shall not be cleaned cathodically or acid cleaned.

3.5 Underplating. Surfaces other than nickel, silver, gold, or platinum shall be nickel plated (or nickel over copper plated as applicable) in accordance with AMSQQN290. An underplating of nickel shall be used when parts are of corrosion or heat resistant steels. The nickel plating shall be of such a thickness that after rhodium plating, the drawing dimensions shall be met. The thickness of the underplating shall not be used in the determination of the rhodium plating as stated herein.

3.6 Thickness. Thickness of the plating shall be in accordance with the requirements of the class specified by the procuring agency (see 1.2 and table I). Unless otherwise specified, the minimum plating thickness shall be as specified on all visible surfaces which can be touched by a ball 0.75 inch in diameter.

Table I. Minimum thickness for rhodium coating classes.

Class of Rhodium Plating	Minimum Thickness	
	(Micrometers)	(Inches)
1	0.0508	0.000002
2	0.2540	0.00001
3	0.5080	0.00002
4	2.5400	0.00010
5	6.3500	0.00025

3.7 Coverage. The plating shall cover all surfaces as stated in 3.6 including corners, recesses, and roots of threads.

3.8 Adhesion. The adhesion of the rhodium plating and any undercoat(s) shall be such that when testing; in accordance with 4.5.2 and examined at a magnification of four diameters, neither the rhodium plating nor any electrodeposited undercoat shall show separation from the basis metal or from each other at their common interfaces.

3.9 Finish. When required for reflective purposes, the plating shall be sufficiently smooth to give a reflectance of at least 75 percent when tested in accordance with 4.5.3 (see 6.2). The reflectance standard shall be as specified by the procuring activity.

3.10 Workmanship. The plated articles shall be of smooth surface and free of cracks, streaks, burns, milkiness, poor coverage, built-up edges, or any other defect which might adversely affect the use or function of the part.

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4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions. Reference shall be made to ISO 9000 for definition of quality assurance terms.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Preproduction sample inspection (see 4.3).

(b) Lot acceptance inspection (see 4.4).

4.3 Preproduction sample inspection. When specified in the contract or order, prior to the production plating of any article, the contractor shall prepare preproduction samples representing the minimum acceptable plating quality to be used in production. The preproduction sample shall re articles of the same basis metal and prepared for plating by similar procedures to those applying to the articles to be plated in production. The preproduction samples selected in accordance with 4.4.2 and 4.4.3 shall be subjected to all examination and tests specified herein. Equivalent substitute specimens may be used in lieu of the actual articles to be plated in production subject to the provisions of 4.4.3.1. Unless otherwise specified in the contract or order, the Government will perform the examinations and test for preproduction acceptance at the contractor's plant. Preproduction samples which do not meet all the requirements of this specification shall be rejected and returned to the contractor. Before submitting another preproduction sample, full particulars concerning previous rejection and the action taken to correct the defects shall be made available to the contracting officer. Articles to be plated under this specification will not be considered for acceptance until approval of the preproduction samples has been obtained.

4.4 Lot acceptance inspection. Lot acceptance samples shall conform with the sampling procedures and tests specified herein.

4.4.1 Lot. A lot shall consist of plated articles of the same class, of approximately the same size and shape, plated under the same conditions by the same contractor and submitted at one time for inspection.

4.4.2 Sampling for visual examination. Unless otherwise specified in the contract or order, samples for visual examination shall be established in accordance with MIL-STD-1916 and conducted in accordance with ANSI/ASQ Z1.4, inspection level III, AQL (acceptable quality level) = 1.0 percent defective.

4.4.2.1 Visual examination. Samples selected in accordance with 4.4.2 shall be examined for compliance with the workmanship requirements (3.10).

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4.4.3 Sampling for thickness, adhesion, finish and embrittlement relief. Unless otherwise specified in the contract or order, samples for thickness, adhesion, finish, and embrittlement relief shall be composed of a minimum of four plated articles selected at random from each lot.

4.4.3.1 Use of equivalent specimens. Unless otherwise specified in the contract or order, equivalent specimens may be included in a lot as substitutes for plated articles that are not suited to the tests specified herein. Equivalent specimens shall be of the same basis metal; shall be prepared for plating in the same manner as the article it is to represent and shall be plated concurrently with the article. Controllable conditions that affect the plating should be replicated as nearly as possible the same for the equivalent specimens as for the represented article. The substitute specimens shall be 1.0 by 4.0 by 0.04 inches.

4.5 Test methods.

4.5.1 Thickness of plating (nondestructive and destructive tests). For nondestructive measuring of plating thickness, procedures in accordance with ASTM B244 (eddy current) may be used. At the option of the supplier, alternate NDT methods including ASTM B499 to measure nonmagnetic coatings on magnetic basis metals, beta radiation back scatter in accordance with ASTM B567, or x-ray fluorescence in accordance with ASTM B568 may be used. For destructive measuring, the microscopic method in accordance with ASTM B487 may be used. For plating thickness less than 0.0001 inch, microscopic, magnetic, or eddy current methods should not be used.

4.5.2 Adhesion test. The samples shall be repeatedly bent through an angle of 180 degrees on a diameter equal to the thickness of the specimen until fracture of the basis metal occurs. Following fracture, it shall not be possible to detach areas of the coating by use of a pointed instrument. If the edge of the ruptured plate can be peeled back or if separation between rhodium or underplating and the basis metal, or any plating from each other can be seen at the point of rupture, when examined at 4 diameters magnification, adhesion is not satisfactory. The formation of cracks that do not result in flaking or peeling of the deposit shall not be considered as nonconforming to this requirement.

4.5.3 Finish test. When the rhodium plated surface is to be tested for reflectivity, a device similar to the one shown in figure 1 shall be used. The angle of reflection and incidence shall be measured at 45 degrees plus or minus 2 degrees. A flat surface shall be used for reflectivity tests.

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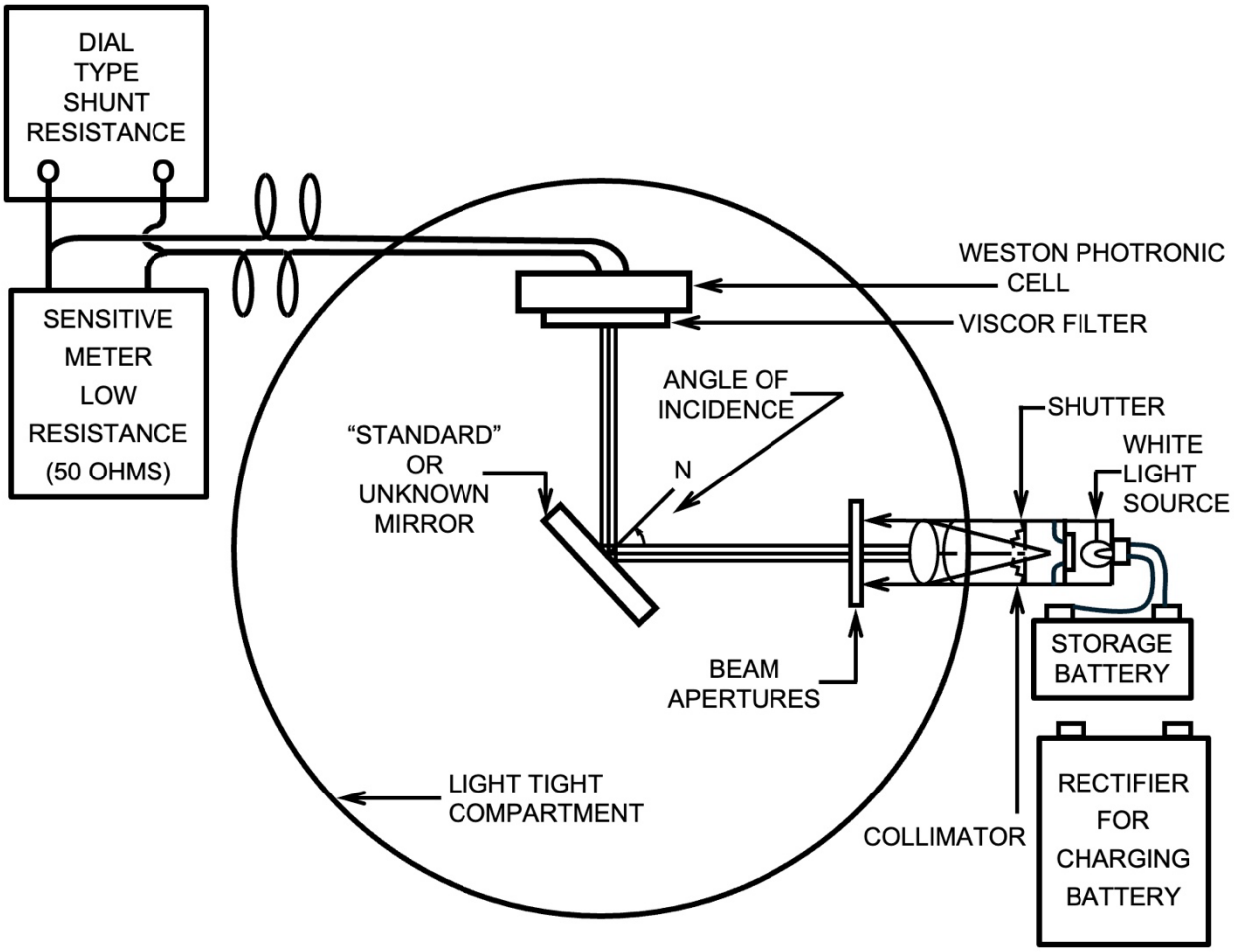


FIGURE I. Finish Test.

4.5.4 Embrittlement relief. Samples of parts as specified in 4.4.3, which will be subjected to sustained tensile load in use, shall be subjected to a sustained tensile load equal to 80 ± 5 percent of the tensile yield strength of the material. The parts shall be held under the load for at least 100 hours and then examined for cracks. The lot shall be rejected if any plated part develops cracks. Parts which require special fixtures, extreme loads to comply with the above requirements, or where maximum design yield load is not known, may be represented by separate specimens. The configuration shall be in accordance with figure 8 of ASTM E8. The notched specimens shall be subjected to a sustained tensile load equal to 75% of the ultimate notch tensile strength of the material. The articles, parts or specimens shall be held under load for at least 200 hours and then examined for cracks or fracture.

4.6 Rejection.

4.6.1 Examination for defects. Any sample unit having one or more defects shall be rejected. If the number of nonconforming sample units in the sample exceeds the acceptance number specified in 4.4.2, the entire lot shall be rejected.

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4.6.2 Thickness, adhesion, finish and embrittlement relief tests. A lot shall be rejected for failure to meet any of the test requirements when tested in accordance with 4.5.

4.7 Retest. Individual plated articles or lots of plated articles rejected as defective may be reprocessed and resubmitted for testing under conditions specified by the procuring agency.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.2 Ordering data. Purchasers should exercise any option offered herein and procurement documents should specify the following:

a. Title, number and date of this specification.
b. Class of plating required (see 1.2).
c. Whether a preproduction sample is required (see 3.1).
d. If surface is required for reflective purposes (see 3.9).
e. Sampling plan and acceptable quality level (see 4.4.2).
f. Preparation for delivery requirements (see 5.1).

6.3 Application of rhodium. The article to be plated should be placed in a plating bath in such a position that the surface to be plated will be completely covered by the solution. The temperature and concentration of the plating bath should be established and accurately controlled together with voltage and current density in order to meet the requirements of this specification and the specified thickness.

6.4 Safety caution. In making up the plating bath, it is important that the acid be added to the water before adding the rhodium concentrate to prevent precipitation of rhodium compound of hydrolysis. Never add water to the concentrated acid, as explosive steaming may occur and throw acid from the container.

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6.5 Handling. Plated parts should be handled with clean gloves and wrapped in neutral tissue paper for temporary storage.

6.6 Supersession data. This specification includes the requirements of Missile Purchase Description MPD-592B dated 31 January 1961 and MIL-R-46085A dated 18 June 1964. Table II gives cross-references between these documents.

Table II. Thickness class cross-reference. 1/

Specification Thickness Class Alignments		
MPD-592B	**MIL-R-46085 (MR)**	**MIL-R-46085A & B**
1	1	1
2	1	---
3	2	2
4	2	---
5	3	3
6	4	4
7	5	5

1/ No change in classification between MIL-R-46085A through MIL-DTL-46085C.

6.7 Subject term (key word) listing.

Cables	Electronics	Plug
Connectors	Electroplating	Precious metals
Contacts	Grounding	Reflector
Electrical	Platinum group metals (PGMs)	Underplating

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - MR
Air Force - 20
Navy - AS

Preparing activity:
Army - MR

(Project MFFP-2025-006)

Review activities:
Army - AR, MI
Air Force - 70, 99

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